

1. NAME OF THE MEDICINAL PRODUCT

Vicombil, coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Retinol 5000 I.U., Ergocalciferol 500 I.U., Tocopherol acetate 10 mg, Thiamine HCl 3 mg, Riboflavin 2 mg, Pyridoxine HCl 3 mg, Cyanocobalamin 1 mcg, Ascorbic acid 75 mg, Nicotinamide 20 mg, Folic acid 0,2 mg, Calcium pantothenate 5 mg (per coated tablet).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Coated tablets.

4. CLINICAL PARTICULARS

4.1. Therapeutic indications

Prophylaxis and treatment of multivitamin deficiencies.

4.2. Posology and method of administration

Posology

Adults and adolescents:

Prophylaxis: 1 tablet daily.

Treatment: 1 tablet, 2 to 4 times daily.

4.3. Contraindications

VICOMBIL is contraindicated in situations of hypervitaminosis and hypersensitivity to one or more vitamins included in its composition.

4.4. Special warning and special precautions for use

Allergic hypersensitivity to one or more vitamins included in VICOMBIL.

4.5. Interactions with other medicinal products and other forms of interaction

Some of the vitamins included in VICOMBIL may have interactions with other drugs. However, if therapeutically adequate, usually the doses of vitamins included in VICOMBIL are not too high to develop such interactions. The administration of cholestyramine may result in decreased absorption of vitamin A and D. Oral administration of neomycin may also result in decreased absorption of vitamin A. Since tendency for additive adverse effects is relevant, patients receiving a retinoid (e.g., etretinate, isotretinoin) should be warned to avoid concomitant use of preparations containing vitamin A. Concurrent administration of thiazide diuretics and pharmacological doses of vitamin D analogues in patients with hypoparathyroidism may result in hypercalcemia. Corticosteroids counteract the effects of vitamin D analogues. Concurrent use of vitamin D analogues and cardiac glycosides may result in cardiac arrhythmias. Vitamin E or one of its metabolites may have anti-vitamin K effects. Pyridoxine HCl reverses the therapeutic effects of levodopa by accelerating peripheral metabolism of levodopa. Absorption of vitamin B₁₂ from the GI tract may be decreased by

aminoglycoside antibiotics, colchicine, extended-release potassium preparations, aminosalicic acid and its salts and anticonvulsants (e.g. phenytoin and phenobarbital). Increased urinary excretion of ascorbic acid and decreased excretion of aspirin occur when the drugs are administered concurrently. Folic acid may increase phenytoin metabolism, resulting in decreased phenytoin serum concentrations and so an increase in seizure frequency may occur in some patients. In epileptic patients, undergoing anticonvulsant therapy, high doses of folic acid may lead, especially in children, to convulsions, thus requiring an increase of the anticonvulsant dose.

4.6. Fertility, pregnancy and lactation

Vicombil may be used during pregnancy and lactation. However, the use of large doses of vitamin A should be avoided generally during normal pregnancy. The use of large doses of pyridoxine during pregnancy has been associated to pyridoxine-dependence seizures in neonates.

4.7. Effects on ability to drive and use machines

Vicombil has no or negligible influence on the ability to drive and use machines.

4.8. Undesirable effects

If the recommended doses are not exceeded side effects will not be observed. Overdosage may develop signs of hypervitaminosis A and D.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected directly to the Pharmaceutical Services, Ministry of Health, CY-1475, www.moh.gov.cy/phs Fax: + 35722608649

4.9. Overdose

Some vitamins included in VICOMBIL (e.g. vitamins A and D) may develop symptoms of hypervitaminosis, if large doses are administered, for long term periods. Therefore the recommended dose should not be exceeded.

Toxicity due to high doses of vitamin A is more common in young children than in adults. Some hours after the intake of vitamin A at a dose approximately of 25.000 units/kg may develop irritability, nausea, diarrhoea, vomiting, vertigo, delirium and coma.

Usually ascorbic acid is nontoxic. However diarrhoea may occur after an oral dose of 1 g or over. Acute or chronic administration of excessive doses of vitamin D analogues may lead to hypervitaminosis D manifested by hypercalcemia. The necessary dose of vitamin D analogue to develop hypervitaminosis varies from individual to individual.

Early symptoms of hypercalcemia may include asthenia, fatigue, somnolence, headache, anorexia, dry mouth, metallic taste, nausea, vomiting, constipation, diarrhoea, vertigo, ataxia, exanthema, hypotonia (in children), muscle pain, bone pain, and irritability. Later, and sometimes more serious, complications of hypercalcemia may include nephrocalcinosis, renal function impairment (resulting in polyuria, nocturia, polydipsia and proteinuria), osteoporosis in adults, growth retardation in children, weight loss, anaemia, calcific conjunctivitis, photophobia, metastatic calcification, pancreatitis, generalised vascular calcification, and seizures.

Treatment of intoxication due to vitamin D analogues consists of the withdrawal of the medicine and of calcium supplementation, use of a low calcium diet, oral or endovenous administration of fluids and, if needed, corticosteroids or other drugs, particularly calcium diuretics (e.g. furosemide

and etacrinic acid) to reduce serum calcium concentrations. Haemodialysis or peritoneal dialysis may also be used. In acute intoxication due to a vitamin D analogue, if intake was recent, gastric lavage or emesis should be induced to prevent absorption.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: 11.3.1.3 Association of vitamins, ATC code: A11B A.

Vitamins are a group of substances with several chemical compositions which may be defined as organic substances that should be supplied in small amounts in diet, to the synthesis, by tissues, of co-factors which are essential to several metabolic reactions. Vitamins may be used in Medicine by two different purposes:

- 1- to treat a specific vitamin deficiency (e.g. ascorbic acid to treat scurvy, or nicotinic acid to treat pellagra);
- 2- to be administered as a multivitamin preparation, to compensate vitamin deficiencies resulting from diet content.

The second type of use may be prophylactic, or, if the multivitamin deficiency already exists, with therapeutic purposes.

VICOMBIL is a multivitamin formulation to be used as supplement when an inadequate intake of vitamins is evident. Its balanced composition according to the daily vitamin requirements as established by the *Dietary Allowances Committee of the Food and Nutrition Board of the USA* (RDA). The use of vitamin supplements is clinically recommended in several circumstances where vitamin deficiencies may take place, being this practice universally accepted.

Retinol plays an important role on the retina function. It is necessary for growth and differentiation of the epithelial tissue, bone growth and embryonal development. The recommended dose as supplement is of approximately 5.000 IU (corresponding to Retinol 1.000 Eq). Daily intake of this dose does not develop vitamin A accumulation or Risk of hypervitaminosis.

Thiamine Hydrochloride acts on the body as thiamine pyrophosphate. This last has functions on the carbohydrate metabolism, as coenzyme in carbohydrate, in the decarboxylation of alfa-ketoglutaric acids and in the utilization of pentose in the hexose monophosphate shunt. It appears to act as modulator of cholinergic transmission.

Thiamine requirements are related with the metabolic rate, being higher when the carbohydrates are the main energy source. Daily supplemental intake should not be less than 1.5 mg, but in several circumstances, it is necessary to increase the amount of this vitamin. Thiamine deficiency may occur in association with an apparently adequate dietary intake, since this vitamin is not stored in the human body in large amounts.

Riboflavine has its functions in the organism as one of two coenzymes. This vitamin plays a fundamental role in the metabolism, as coenzyme in a wide range of respiratory flavoproteins, some of which containing metals. The daily supplementation should not be less than 1.6 mg.

Pyridoxine hydrochloride is converted in the human body to three active forms of the vitamin: pyridoxine, pyridoxal and pyridoxamine. This vitamin plays an important role in the metabolism, as coenzyme in a wide variety of reactions in intermediary metabolism of amino acids, including decarboxylation and transamination, as well as to the enzymatic metabolic steps of tryptophan and hydroxy-amino acids. It is recommended a supplemental intake not less than 2.2 mg.

Cyanocobalamin is converted to two active coenzymes: methylcobalamin and deoxyadenosylcobalamin. These active coenzymes are essential to growth and cell multiplication. Deoxyadenosylcobalamin is a co-factor to the mitochondrial mutase enzyme that catalyses L-methyl-malonil-CoA isomerization to succinyl CoA, an important reaction both for carbohydrates metabolism and lipids. Methylcobalamin supports the methionine synthetase reaction, which is essential to the normal folate metabolism. Interaction folate-cobalamin is essential to the normal synthesis of purines and pyrimidines and therefore to ADN. It is also responsible for the maintenance of a certain amount of methylation reactions. When vitamin B₁₂ concentrations decrease, folate remains imprisoned as methyl-tetrafolate, leading to a functional deficiency of other vital and intracellular forms of folic acid.

An additional dose of 1mcg of B₁₂ is commonly adequate. When higher doses are needed, a preparation containing exclusively B₁₂ should be administered. IM route should be used in case of pernicious anaemia, a disease for which a multivitamin preparation such as VICOMBIL is not adequate.

Ascorbic acid (Vitamin C) enters in various biochemical reactions, most of them involving oxidation. It is a strong reducing agent and participates in oxidation-reduction cellular reactions. It appears to be particularly important to bone, teeth and collagen formation. In the adrenal cortex there are high amounts of vitamin C from where it can be released by ACTH intervention. The most important sources of vitamin C are fruits and fresh green vegetables. It is usually recommended a daily dose not below 65 mg.

Ergocalciferol (vitamin D₂) is a vitamin D stable form which is active when it is administered when vitamin D is mandatory. Ergocalciferol is transferred to the liver where it is converted to calcifediol, which is then transferred to the kidney and converted to calcitriol, the most active form of vitamin D. However, in medicinal formulations, ergocalciferol is the choice because in opposition to calcitriol, it is chemically stable. Vitamin D is a regulator of calcium homeostasis. The two most important mechanisms through which vitamin D acts to maintain normal serum concentrations of calcium and phosphate in plasma, are the absorption facilitation of both at the intestinal level and promotion of its mobilization through the bone. Also, vitamin D₂ acts upon the kidney leading to a retention of calcium and phosphorous due to tubular resorption.

Doses under 5.000IU (12.5 mcg) or over 50.000 IU should not be administered, according to clinical indication (prophylaxis or treatment).

Nicotinamide (vitamin PP) occurs in the liver, meat and yeasts and may be synthesised by intestinal bacteria. It is a component of coenzymes and nucleotides and is present in live cells.

When associated to specific proteins of oxidation-reduction systems, these coenzymes, accept hydrogen atoms which further transfer to other enzymes. This vitamin is, therefore, necessary to several reactions that are essential to cells life.

Folic acid is a very important enzyme to the cellular metabolism. After absorption, the pteroylglutamic acid is rapidly reduced in positions 5,6, 7 and 8, giving birth to tetra-hydrofolic acid which acts as a receptor to a set of units of one-carbon atom. They link to positions 5 or 10 originating active forms of coenzyme, each of them having a specific metabolic role.

The active form of **calcium pantothenate** is the coenzyme A. It serves as co-factor in several reactions catalyzed enzymatically and involving acetylic groups transfer. These reactions are important in oxidative metabolism of carbohydrates, in glycogenesis, in the synthesis and degradation of fat acids, in the synthesis of sterols, of steroid hormones and porphyrines. Calcium pantothenate may be synthesized by intestinal bacteria.

5.2. Pharmacokinetic properties

All vitamins in VICOMBIL are well absorbed by digestive tract. Once absorbed they are more or less stored and metabolized. However vitamin B₁₂ is rapidly absorbed if there does not exist deficiency of the intrinsic factor or ileum disease.

Vitamin A is stored in the liver in several stores, from which other organs are supplied. Vitamin A concentration in the liver is about 100 mcg/g. Retinol is partially conjugated, forming a beta-glucuronid conjugate that enters in the entero-hepatic circulation and is oxidised to retinal and retinoic acid.

Vitamin B₁, which has a maximum absorption limit by the digestive tract of 8-15mg/day, is completely degraded in tissues if the administered amount does not exceed 1 mg/day. When that amount exceeds 1 mg, organic stores are saturated. The remaining appears in urine as unchanged thiamine or pyrimidine. Only 9% of Vitamin B₂ appears in urine when the intake is near the daily requirements. The more the intake exceeds these requirements, the more of unchanged Vitamin B₂ is excreted.

Also, nicotinamide is distributed to tissues. When it is administered at therapeutic doses, only small amounts appear in urine. If those doses increase larger amounts appear progressively in urine.

As with other vitamins, Vitamin B₆ is well absorbed orally, and is excreted, mainly as 4-pyridoxic acid.

Calcium pantothenate is well absorbed and is excreted unchanged. Vitamin B₁₂ deficiencies reflect, in general, a reduction on the capacity of absorption of the gastrointestinal mucosa. Once absorbed Vitamin B₁₂ binds to transcobalamin II, a plasmatic globulin that transport it to the tissues. The liver is the major store. More than 90% of Vitamin B₁₂ in the body (1-10 mg) is in that organ.

When folic acid is absorbed, it is rapidly converted to CH₃ H₄ Pteglu₁. Folates are then stored in the cells as glutamates.

After Vitamin C is absorbed, it is distributed to every cell in the body. A daily dose of 5-10 mg leads to an organic store of 6000-1.000 mg ascorbic acid. If daily intake reaches 200 mg, the stores increase to 2.500 mg, with plasmatic levels of 2 mg/dl. Since the renal threshold is 1.5 mg/dl, the kidney excretes the amounts exceeding 100 mg intake.

Vitamin E enters in the blood flow helped by the lymph. It is distributed to all tissues. Seventy-eight per cent of vitamin E is excreted in the liver.

After Vitamin D absorption, it circulates in the blood bound to a protein. The main excretion route of Vitamin D is the bile.

5.3. Preclinical safety data

The safety of use of the different vitamins in VICOMBIL is well established: The hydrosoluble ones are practically nontoxic, except for the cases of susceptibility to vitamins B₁ and B₂ and the possibility of folic acid, when administered alone in case of pernicious anaemia, to hide a haematological condition and allow the development of haematological complications.

As for the liposoluble vitamins, especially A and D, they may, if high doses are administered, be toxic to Man, leading to hypervitaminosis, especially in children. Therefore, in some adults taking doses higher than 25.000/day of vitamin A, that is, 5 times more the daily recommended doses, some hypervitaminosis diagnosis occurred.

Also, the administration of ergocalciferol, acute or chronic (1.25 to 2.5 mg/day) led to hypervitaminosis D, manifested by hypercalcemia.

For prophylactic purposes the multivitaminic preparation may contain 50 to 100% of RDA, and for therapeutic purposes 3-5 times the RDA, except for vitamin D and folic acid which should not exceed the RDA.

Both the quantity per dose unit and the posology established for VICOMBIL meet these exact requirements.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Calcium phosphate, polyvinyl pyrrolidone, microcrystalline cellulose, magnesium stearate, talc, methyl hydroxypropyl cellulose, Opaspray, diethyl phthalate, ethyl cellulose.

6.2. Incompatibilities

Not applicable.

6.3. Shelf-life

3 years.

6.4. Special precautions for storage

Do not store above 25°C.

6.5. Nature and contents of container

Carton box containing 60 tablets in blister Alu/Alu and a leaflet information.

6.6. Special precautions for disposal

No special requirements for disposal.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MARKETING AUTHORIZATION HOLDER

BIAL – Portela & C^a, S.A.
À Av. da Siderurgia Nacional
4745 - 457 S. Mamede do Coronado
Portugal

8. MARKETING AUTHORISATION NUMBER(S)

18570

9. DATE OF FIRST MARKETING AUTHORISATION/RENEWAL OF THE AUTHORISATION

Date of first authorisation: 3 February 2000

Date of latest renewal: 27 May 2011

10. DATE OF REVISION OF THE TEXT

12/11/2020